Vision-Based Precision Pose Estimation For Autonomous Formation Flying



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Problem statement

- Leader plane, follower plane
- Follower plane views leader through camera
- Camera is gimballed at leader (leader is always in the center of the frame)
- Must predict the pose of the leader plane (x, y, z and roll)

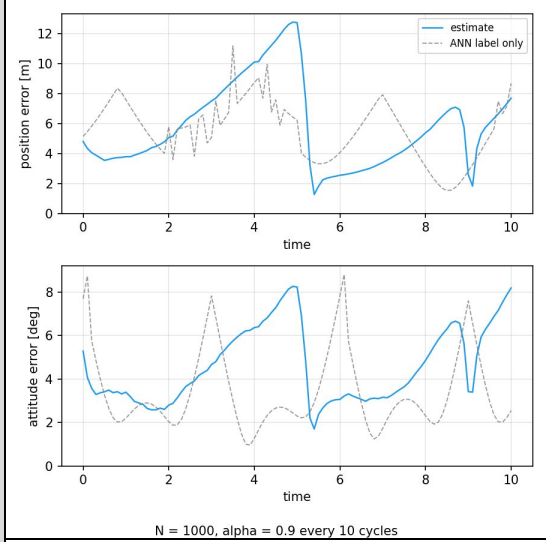
Parameter	Axis description	Convention used for positive value
x	Leader to follower	In front of the leader
y	Horizontal of the follower	Right side of the follower
z	Vertical of the follower	Below the follower
Roll (phi)	Tilt along the nose-to-tail axis	Right wing tips downwards

Dataset

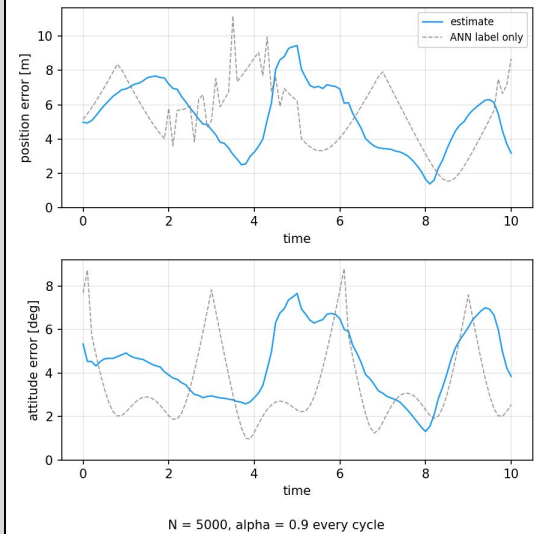
- Synthetically generated
- Simplified F-16 mesh, picked 14 points on it. (u, v, occ)
- Figure out which features are occluded via ray casting
- Labels:
  - In paper's approach: 1 of 4800 pose classes
  - In our approach: Values for x, y, z and roll
- Size: 1M records

Paper's approach

1. Generate dataset as given in the previous slide
2. Train the ANN to get a discrete pose
   - ANN specs: 42 (ReLU) → 100 (ReLU)→ 100 (ReLU)→ 4800 (softmax)
3. Apply a particle filter to converge on the real pose values from the discrete pose
4. Test on a scenario where the follower plane closes in on the leader for 10s



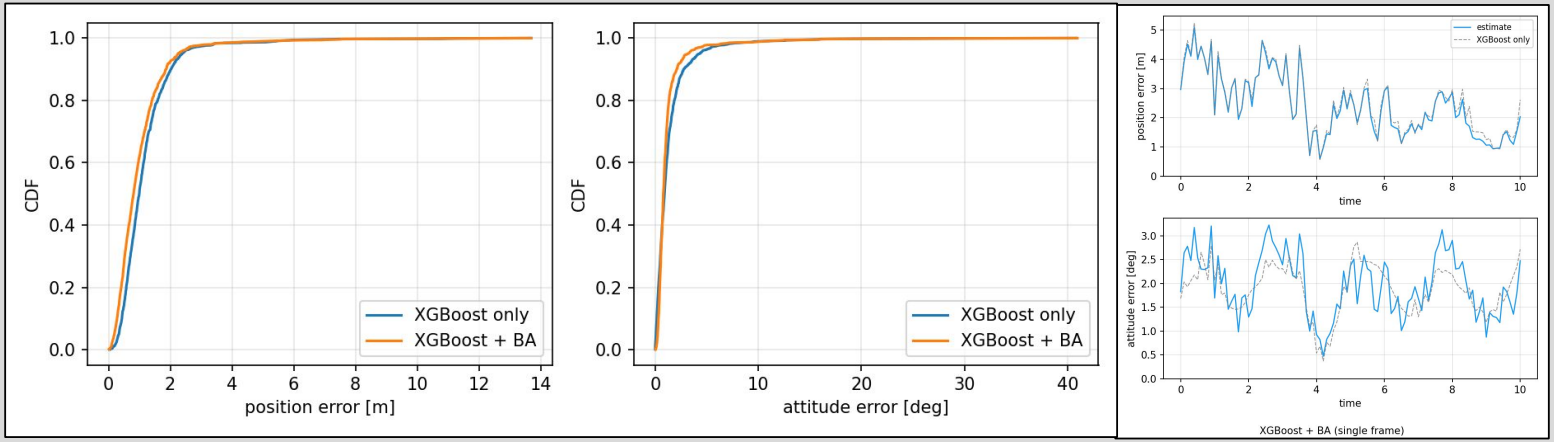
N = 1000, alpha = 0.9 every 10 cycles



N = 5000, alpha = 0.9 every cycle

Our approach

1. Generate dataset as given in the previous slide
2. Train XGBoost to get a continuous value of x, y, z and roll
3. Apply bundle adjustment to refine the answer of XGBoost
4. Test on a scenario where the follower plane closes in on the leader for 10s



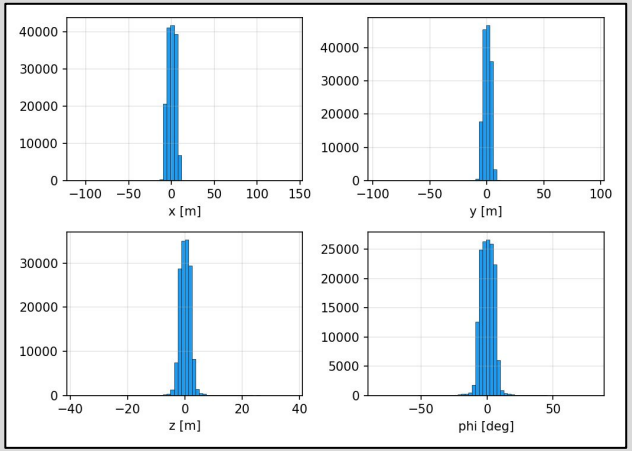
Comparison and key findings

ANN + Particle filter	XGBoost + bundle adjustment
Classification (pose is binned into 4800 discrete tuples for x, y, z and roll)	Regression (pose parameters are continuous)
Error is greater due to binning	Error is reduced because regression was used
Particle filter (informed by physics) smoothens the ANN error curve but still follows it	Bundle adjustment follows the error curve of the XGBoost and does not smoothen at all

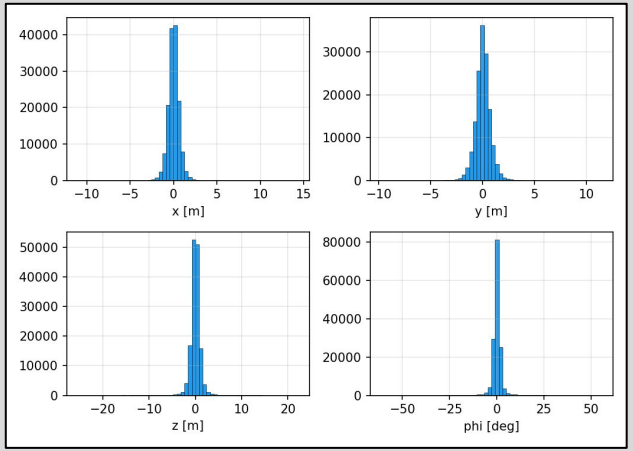
- Bundle adjustment did not help much with the accuracy of XGBoost.
- Neither the particle filter or bundle adjustment were able to substantially reduce the error of the previous model.

Results

- Findings from the paper was reproduced successfully (particle filter did not fix the errors of the ANN)
- Second pipeline with XGBoost reduced the error of the first



ANN



XGBoost

Thank you!